MIL-L-3061B(OS)
13 June 1969
SUPERSEDING
MIL-L-3061A(NOrd)
1 June 1954

MILITARY SPECIFICATION

LECITHIN (FOR USE IN EXPLOSIVES)

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

* 1.1 This specification covers one grade of vegetable lecithin extracted from soybeans, for use in explosives.

2. APPLICABLE

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 6810

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American Society for Testing and Materials Standards

D1639-61 Acid Value of Organic Coating Material

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

American Oil Chemists Society

A.O.C.S. Tentative Method Ja 2-46 - Moisture

(Application for copies should be addressed to the American Oil Chemists Society, 35 East Wacker Drive, Chicago, Illinois 60601.)

3. REQUIREMENTS

* 3.1 First Article inspection - The lecithin furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein. (See 4.3 and 6.2.1).

* 3.2 Chemical and physical properties - Lecithin shall be a plastic grade and shall conform to the chemical and physical properties of Table I when tested as specified therein.

TABLE I

CHEMICAL AND PHYSICAL PROPERTIES

PROPERTY	VALUE		TEST PARAGRAPH
	MINIMUM	MAXIMUM	
Moisture, percent	-	1.0	4.5.1
Benzene insoluble matter, percent	-	0.1	4.5.2
Acid number	-	24	4.5.3
Acetone insoluble matter, percent	68	-	4.5.4
Lecithin, percent	19.0	-	4.5.5

3.3 Workmanship - Lecithin shall be homogeneous, free from foreign matter, lumps, and any oil separation on top of the material.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The inspection of the lecithin shall be classified as follows:

a. First article inspection (4.3)
b. Quality conformance inspection (4.4)

4.3 First article inspection - First article inspection shall consist of all the requirements and of all the tests and inspections of this specification, and any additional inspections required by the contract or purchase order.

4.3.1 Sampling for first article inspection - Unless otherwise specified (see 6.2.1), as soon as practicable after award of a contract or order, the supplier shall furnish a first article sample to determine conformance with this specification. The first article sample shall consist of two pounds of lecithin manufactured by the procedure and processes and at the same location proposed by the supplier for the execution of the contract. The sample shall be subjected to the first article inspection specified in 4.3. Failure of the lecithin to meet all requirements shall result in rejection of the first article sample.

4.3.1.1 When the first article sample has been approved, the supplier will be notified and will be authorized by the procuring activity to proceed with production. Any production started before such approval shall be at the supplier's risk. The first article sample accepted will be applied as part of the quantity specified by the contract or order.

4.4 Quality conformance inspection - The quality conformance inspection shall consist of all the examinations (4.4.1.3 and 4.4.2) and tests of this specification with the exception of the percent lecithin requirement (Table I and 4.5.5).

* 4.4.1 Sampling -

* 4.4.1.1 Inspection lot - A lot shall consist of the lecithin produced by one manufacturer at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

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* 4.4.1.2 Samples for test - A random sample of filled containers shall be selected in accordance with MIL-STD-105. The Inspection Level shall be I, and the acceptance number zero. Sufficient quantity of sample shall be removed from each selected container and mixed thoroughly to form a composite sample of approximately two pounds. The composite sample shall be placed in an air-tight container and subjected to the examinations of 3.3 and individual tests of 4.5, except lecithin (4.5.5).

* 4.4.1.3 Sample for examination of filled containers - A random sample of filled containers shall be selected from each inspection lot in accordance with MIL-STD-105. The Inspection Level shall be II, and the Acceptance Quality Level (AQL), 2.5 percent defective. Samples selected shall be examined to verify compliance with all requirements of this specification in regard to fill, closure, marking and other requirements not involving tests.

4.4.2 Examination for oil separation - Inverting the open containers selected in 4.4.1.3 shall cause no loss of liquid material.

4.5 Test methods -

* 4.5.1 Moisture - The moisture content shall be determined in accordance with American Oil Chemists Society (A.O.C.S.) Tentative Method Ja2-46.

4.5.2 Benzene-insoluble matter - Transfer an accurately weighed portion of approximately 10 gm of the sample to a beaker containing 100 ml of benzene. Heat the beaker and contents on a steam bath with occasional stirring until the soluble material has been completely extracted. Transfer the insoluble matter quantitatively to a weighed Gooch crucible. Wash the crucible and contents with 100 ml of benzene and aspirate until dry. Dry in an oven at 100 to 105°C for 1 hour, cool in a desiccator and weigh. Calculate the increase in weight as percent benzene-insoluble matter.

4.5.3 Acid number - Transfer an accurately weighed portion of approximately 10 gm of sample to a 250-ml Erlenmeyer flask containing 50 ml of benzene. Heat the flask and contents on the steam bath with occasional shaking until the sample completely dissolves. Cool the contents of the flask to room temperature, add 50 ml of cold distilled water, and shake. Add 1 ml of 0.2-percent phenolphthalein indicator solution to the flask. Titrate the free acid present with N/10 sodium hydroxide solution. The presence of a faint pink coloration in the aqueous layer which persists after separation of the two layers shall be considered the end point. Make a blank determination using the same amount of benzene, distilled water, and phenolphthalein indicator solution as used for the sample determination. Calculate the acid number as follows:

$$\text{Acid number (mg. of KOH per gm.)} = \frac{56.1\,(V-v)N}{W}$$

where

V = ml of NaOH solution required to titrate the acid in the sample
v = ml of NaOH solution required to titrate the acid in the blank
W = wt of sample in grams
N = normality of NaOH solution

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4.5.4 Acetone insoluble matter - Warm the sample to soften the product (not over 60°C) and then mix thoroughly. Accurately weigh 2 grams of well mixed sample into a centrifuge tube which has been previously tared with a stirring rod. Add 3 ml of petroleum ether and stir well to dissolve soluble matter. Thorough mixing is essential, and will require about 15 minutes. Add exactly 15 ml of acetone from a buret, mix thoroughly, and place the tube in a dish containing ice water. Add acetone, which has been previously chilled to 0° to 5°C, to the 50 ml mark on the tube, stirring during the addition. Place in an ice-water bath at 0°C for 15 minutes, Centrifuge for 10 minutes at a speed sufficient to clear the solution during this period and then decant the acetone into a 250-ml beaker which has been previously dried at 105°C, desiccated and weighed. Refill the centrifuge tube to the 50 ml mark with acetone, stir well and repeat as above. Decant the acetone into the 250-ml beaker. Evaporate the acetone on a water bath. A stream of clear dry air may be used to facilitate the removal of the solvent. Dry in an oven at 105 ± 2°C for one hour, cool to room temperature in a desiccator and weigh the acetone soluble matter.

$$\text{Acetone soluble matter, percent} = \frac{\text{Weight of acetone extract X 100}}{\text{Weight of sample}}$$

$$\text{Acetone insoluble matter, percent} = 100 - (A+B+C)$$

A = Percent acetone soluble matter
B = Percent moisture, from 4.5.1
C = Percent benzene insolubles, from 4.5.2

4.5.6 Lecithin - Transfer an accurately weighed sample of 2 to 3 gm of material to a 150-ml beaker. Add 25 ml of alcohol and 25 ml of (1:1) hydrochloric acid solution. Cover the beaker with a watch glass and place the beaker and contents on a steam bath for 16 hours. Remove the watch glass and evaporate the solution to about 25 ml. Transfer the solution in the beaker to a 200-ml glass-stoppered separatory funnel and rinse the beaker with 50 ml of water. Transfer the rinsings to the separatory funnel. Add 50 ml of ether to the solution, stopper the funnel and shake thoroughly. Allow the aqueous layer to separate completely and draw off the lower layer into a 250-ml beaker. Wash the ether layer twice with 20-ml portions of distilled water and transfer the washings to the beaker. Place the beaker and contents on a steam bath in order to remove the dissolved ether. Add 30-percent sodium hydroxide solution until just neutral to litmus, then add 1 ml of concentrated hydrochloric acid solution. Filter the solution, transfer the filtrate to a 200-ml volumetric flask and dilute to mark with distilled water. Transfer a 25-ml aliquot of the solution to a 150-ml beaker. Add 25-ml of distilled water and 20-ml of a freshly prepared, filtered, saturated solution of Reinecke salt (NH_4 (Cr $(NH_3)_2$ $(SCN)_4)\cdot H_2O$) in distilled water. Allow the precipitate which forms to stand for 15 minutes and filter through a fine porosity fritted glass crucible by means of suction. Wash the precipitate twice with the cold distilled water and once with absolute alcohol. Dissolve the precipitate in the crucible using 15-ml of acetone and transfer to a 50-ml volumetric flask. Wash the crucible three times with 10-ml portions of acetone, Dilute to mark with acetone. Prepare standards by dissolving 0.500 gm of choline chloride, which has been recrystallized from absolute alcohol and dried in an oven at 100 to

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150°C for 16 hours, in distilled water and dilute to 500 ml in a volumetric flask. Transfer by means of pipets or a buret, 5-, 10-, 15-, and 20-ml portions of this standard solution to 150-ml beakers. Add 25 ml of distilled water and 20 ml of the Reinecke salt solution and proceed as directed for the sample. Compare the sample with the standards by means of photoelectric colorimeter using a filter which transmits light in the range of 500 to 550 u, or any other instrument which will give similar results. Calculate the percent lecithin as follows:

$$\text{Percent lecithin} = \frac{557.2A}{W}$$

where

A = gm of choline chloride present in sample determined by comparison with standards
W = gm of sample contained in the aliquot.

* 4.6 Rejection criteria - If a sample fails to meet any of the test or examination requirements of this specification, or if the sample of filled containers exceeds the specified AQL (see 4.4.1.3), the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing -

5.1.1 Levels A and B - Not applicable

5.1.2 Level C - The lecithin shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery specified in the contract or purchase order.

5.2 Marking - In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

* 6.1 Intended use - Lecithin furnished under the requirements of this specification is used in formulations to emulsify and desensitize explosives.

* 6.2 Ordering data - Procurement documents should specify the following:

a. Title, number and date of this specification

b. Quantity in pounds.

6.2.1 Contracts or orders should specify the following provisions for first article inspection.

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6.2.1.1 Whether first article inspection is required - When a contractor is in continuous production from contract to contract, consideration should be given to waive the first article inspection. If first article inspection is required, indicate:

a. Where the first article inspection is to be conducted (at the supplier's or Government plant).

b. That approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of this specification and contract.

* 6.3 Batch - A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

* 6.4 Supersession data - The lecithin furnished under this specification may be completely interchangeable with and substitutable for the lecithin procured under the previous issue, MIL-L-3061A(NOrd) dated 1 June 1954.

* 6.5 Changes from previous issue - The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Project No. 6810-N829

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wo-ding:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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